PILOT: Inter-individual Differences in Susceptibility to Cyber-manipulation: A Multi-agent Simulation Approach with High-dimension State Space of Political Opinions

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April 5, 2026

Abstract

Understanding which individual characteristics amplify susceptibility to adversarial opinion manipulation is a core question in cognitive security research, yet real-world data combining structured profiles, diverse attack vectors, and repeated opinion measurements is difficult to obtain and ethically constrained. We address this gap using an ontology-driven multi-agent simulation pipeline in which 100 synthetic pseudoprofiles are crossed with four attack vectors spanning distinct persuasion mechanisms—cognitive reframing, emotional appeal, false consensus, and authority-heuristic exploitation—and four political-opinion leaves in a full factorial design ($n = 1{,}600$ scenarios). The primary outcome, adversarial effectivity, measures signed opinion movement in the direction of an adversarially defined goal; mean adversarial effectivity was -14.49 ($SD = 44.40$), with only 32.6% of scenarios producing movement in the intended direction, consistent with pervasive resistance and backfire. Intraclass correlation for absolute shift was ICC(1) = 0.052, indicating that attack–opinion context explains an order of magnitude more variance than stable profile characteristics. In the multivariate profile-moderator model corrected for false discovery, Conscientiousness was the only significant predictor ($b = -3.26$, $SE = 1.14$, $q_{\mathrm{FDR}} = .030$), with higher self-discipline associated with reduced susceptibility across opinion leaves. A leave-one-group-out hierarchical decomposition assigned 23.1% of relative importance to Conscientiousness alone. These results validate a factorial simulation architecture for studying heterogeneity in susceptibility to adversarial persuasion and provide methodological infrastructure for comparison with human experimental data.

1 Introduction

Political influence operations increasingly exploit the gap between information volume and cognitive processing capacity. Rather than relying on technical intrusion, they operate through selective framing, fabricated consensus, pseudo-expert endorsement, and identity-congruent narrative design—mechanisms that shape the conditions of political judgement rather than the content of specific beliefs (Bennett & Livingston, 2018; Lazer et al., 2018; Lewandowsky et al., 2017; Pennycook & Rand, 2019, 2021; Vosoughi et al., 2018). The core question this raises is not simply whether content is false, but whether adversarial exposure predictably shifts opinions in directions that serve strategic goals, and whether that susceptibility varies systematically across individuals.

This question has been increasingly framed in security policy discussions under the term cognitive warfare, recognizing that human attention, trust, and decision-making have become operationally relevant terrain in contemporary conflict and competition (Johns Hopkins University & Imperial College London, 2021; Hung & Hung, 2022; Paulauskas, 2024). We adopt

cognitive sovereignty—the capacity for self-directed political belief formation in the face of adversarial manipulation—as the substantive construct motivating the study design. Susceptibility is then defined as the degree to which post-exposure opinions move in directions intended by a hypothetical adversarial operator, rather than as simple opinion change magnitude.

The methodological approach is grounded in recent demonstrations that large language model (LLM) agents can produce socially and demographically conditioned behavior that approximates human response distributions at the aggregate level (Argyle et al., 2023; Park et al., 2023). These findings support using LLM-generated synthetic profiles as tractable proxies for exploring how individual-difference dimensions interact with adversarial exposure, provided results are interpreted as properties of the simulation rather than direct claims about human populations.

A critical design question is why simulation rather than direct profile-to-score elicitation. Prompting an LLM to rate a profile's vulnerability produces a scalar impression that compresses all information into a single opaque judgement. The simulation pipeline produces something fundamentally different: a conditional response surface that links each profile to a structured outcome panel across attack types, opinion domains, and scoring rules. This yields three specific methodological advantages. First, susceptibility is estimated as a conditional, task-dependent property—the same profile may be exploitable for one issue framing or attack mechanism but not another, and simulation preserves that structure explicitly. Second, repeated indicators across tasks constitute a measurement system: internal consistency, residual covariance, ICC, and SEM paths can be computed and audited, whereas a single elicited score provides no psychometric decomposition. Third, the staged pipeline—baseline elicitation, exposure generation, post-exposure assessment, audit—externalizes the assumptions that determine how profile features map to outcomes, making each link inspectable and comparable to human experimental data independently rather than remaining hidden inside a one-step inference.

The present pilot extends the single-attack design of earlier runs to a full factorial layout, introducing adversarial effectivity as a directional primary outcome and a battery of modern statistical methods including intraclass correlation for variance partitioning, Benjamini–Hochberg FDR correction for the moderator battery, and a leave-one-group-out hierarchical R^2 decomposition for the susceptibility index. The study addresses the question: *Which inter-individual differences moderate adversarial effectivity across repeated political-opinion and attack-vector combinations?*

2 Materials and Methods

2.1 Ontology-Constrained Design

Run 8 uses three hierarchical ontologies—PROFILE, ATTACK, and OPINION—each stored as a typed tree but estimated at the leaf level. PROFILE combines continuous leaves (age; Big Five personality facets expressed as percentile scores) with categorical leaves (sex). OPINION samples four leaf nodes from the domain `Defense_and_National_Security`: Alliance Commitment Support, Civil Defense Preparedness Support, Domestic Security Powers Expansion Support, and Strategic Autonomy Support.

ATTACK contributes four leaf nodes spanning distinct persuasion mechanism families:

1. `Social_Media_Misinformation > Misleading_Narrative_Framing` — selective reframing exploiting cognitive heuristics;

2. `Manipulative_Persuasion > Fear_Appeal_Scapegoating_Post` — emotionally charged content activating threat-based processing;
3. `Coordinated_Amplification > Astroturf_Comment_Wave` — synthetic grassroots consensus exploiting social-proof;
4. `Manipulative_Persuasion > Pseudo_Expert_Authority_Cue` — fabricated expert endorsement exploiting authority heuristics.

The design is a complete factorial: each of 100 pseudoprofiles is crossed with all 4 attack leaves and all 4 opinion leaves, producing $100 \times 4 \times 4 = 1{,}600$ attacked scenarios. All rows are attacked; there is no no-attack control condition, consistent with the moderation-under-exposure estimand.

2.2 Baseline, Exposure, and Post-Exposure Assessment

For each profile i, attack a, and opinion leaf k, the pipeline elicits a baseline opinion and a post-exposure opinion on a signed integer scale from $-1{,}000$ to $+1{,}000$. Exposure texts are generated as platform-native manipulative messages tailored to the specific attack–opinion combination, then audited for realism. Post-exposure responses are audited for coherence and boundedness. Every analyzed row has passed a structured review layer designed to detect implausible reversals, coarse rounding, and issue-incoherent response shifts.

2.3 Adversarial Effectivity Outcome

Let $d_k \in \{-1, +1\}$ denote the pre-specified adversarial goal direction for opinion leaf k. The signed shift for profile i under attack a on leaf k is

$$\Delta_{iak} = \text{post}_{iak} - \text{baseline}_{iak}, \tag{1}$$

and the primary outcome, adversarial effectivity, is

$$\text{AE}_{iak} = \Delta_{iak} \times d_k. \tag{2}$$

Positive AE means the attack moved the opinion in the adversarially intended direction; negative values indicate backfire or resistance. The undirected absolute shift $|\Delta_{iak}|$ is retained as a secondary outcome for variance-partitioning purposes.

2.4 Profile-Level Susceptibility Index

The repeated-outcome SEM estimates profile-level path coefficients from a four-indicator model in which each indicator is the adversarial effectivity for one opinion leaf, averaged across attack vectors:

$$\overline{\text{AE}}_{i \cdot k} = \alpha_k + X_i \beta_k + C_i \gamma_k + \varepsilon_{ik}, \tag{3}$$

where X_i is the profile feature vector and C_i contains covariates. Averaging across attacks is necessary because 16 indicators (4 attacks $\times$ 4 opinions) would oversaturate the model for 100 profiles; attack-specific heterogeneity is examined separately.

The post hoc susceptibility index is derived from target-conditional ridge regression (Hoerl & Kennard, 1970) models fit across all 16 attack–opinion tasks:

$$S_i^*(\mathcal{T}) = \text{PctRank}\left(\sum_{t\in\mathcal{T}} w_t \hat{A}_{it}\right), \quad w_t \propto \frac{n_t}{\text{CV-MSE}_t}. \tag{4}$$

This index is descriptive: it summarizes model-implied profile propensity toward larger adversarial effectivity while shrinking unstable coefficient combinations.

2.5 Statistical Analysis Plan

Intraclass correlation. ICC(1) (Shrout & Fleiss, 1979) partitions total variance into between-profile and within-profile (attack × opinion) components, establishing the ceiling for profile-level moderation effects.

Repeated-outcome SEM. A profile-level path model simultaneously regresses four attack-averaged opinion indicators on profile moderators with freely estimated residual covariances, preserving opinion-specific moderation patterns.

Multiple-comparison correction. Moderator p-values are adjusted using the Benjamini–Hochberg FDR procedure (Benjamini & Hochberg, 1995).

Hierarchical importance decomposition. A leave-one-group-out cross-validated R^2 decomposition quantifies the marginal predictive contribution of demographic and Big Five feature groups to the susceptibility index. Relative importance is $|\text{marginal } R_g^2| / \sum_g |\text{marginal } R_g^2| \times 100$.

3 Results

3.1 Descriptive Overview

Run 8 generated 1,600 attacked scenarios across 100 profiles, 4 attack vectors, and 4 opinion leaves. Mean absolute shift was 43.16 ($SD = 17.86$), mean signed shift was 29.48 ($SD = 36.23$), and mean adversarial effectivity was -14.49 ($SD = 44.40$). Only 32.6% of scenarios produced positive adversarial effectivity: the majority of post-exposure opinions moved against the intended adversarial direction. Mean attack realism was 0.71 and mean post-exposure plausibility was 0.74. The study design and factorial structure are summarized in Figure 1; Table 1 reports full configuration parameters.

3.2 Variance Partitioning

ICC(1) for absolute shift was 0.052 ($\sigma^2_{\text{between}} = 16.62$, $\sigma^2_{\text{within}} = 302.81$); for signed shift, 0.021; for adversarial effectivity, ≈ 0.000. Fewer than 6% of the variance in absolute shift is attributable to stable between-profile differences; the remainder is driven by which attack was applied to which opinion leaf. This variance structure sets a hard ceiling on profile-level moderation: even a perfect profile moderator could explain at most 5–6% of total outcome variance.

3.3 Attack-Specific and Opinion-Specific Patterns

Attack vectors differed meaningfully in adversarial effectivity. Misleading Narrative Framing was most effective (mean AE = -11.98, 35.8% positive), followed by Fear Appeal Scapegoating

(AE = −13.14, 35.5%), Astroturf Comment Wave (AE = −15.80, 31.0%), and Pseudo Expert Authority Cue (AE = −17.05, 28.2%). The cognitive-framing vector thus outperformed the authority-heuristic vector by approximately 5 points on the adversarial effectivity scale.

Opinion leaves showed larger and directionally asymmetric divergence. Strategic Autonomy Support was the only leaf with positive mean adversarial effectivity (AE = +29.98): attacks consistently moved this opinion toward the intended adversarial pole. Civil Defense Preparedness (AE = −36.64) and Domestic Security Powers Expansion (AE = −39.31) showed strong resistance, with most post-exposure opinions moving against the intended direction. Figure 2 shows the absolute-shift distribution by opinion leaf; Supplementary Figure S4 shows the attack-vector comparison panel.

3.4 Structural Equation Model

The repeated-outcome SEM converged with CFI = 1.000 and RMSEA = 0.000 ($\chi^2 = 0.144$, $p = 1.000$, $n = 100$). Fit indices should be interpreted cautiously in a four-indicator, 100-observation model; the near-perfect fit reflects the limited degrees of freedom rather than independent validation. The four opinion indicators are adversarial deltas averaged across attack vectors; indicator-level attack heterogeneity is examined in the supplementary materials.

The strongest and most consistent leaf-specific path was from Conscientiousness to Alliance Commitment ($b = -6.76$, $SE = 2.45$, $p = .006$) and Civil Defense Preparedness ($b = -4.54$, $SE = 1.83$, $p = .013$); both indicate that higher Conscientiousness is associated with reduced adversarial opinion shift. The path from Baseline Absolute Score to Domestic Security ($b = -6.53$, $SE = 1.44$, $p < .001$) reflects a ceiling effect: profiles with strong prior security convictions show smaller absolute movement regardless of attack. Exposure Quality predicted Strategic Autonomy ($b = 4.80$, $SE = 1.85$, $p = .010$), suggesting that this particular opinion leaf responds to attack quality rather than profile characteristics. Full SEM path coefficients are reported in Supplementary Table S5 and illustrated in Figure 4.

3.5 Profile Moderator Analysis

Table 3 and Supplementary Table S2 report the multivariate and univariate moderator comparison with FDR-corrected q-values. In the multivariate model, Conscientiousness was the sole predictor surviving FDR correction ($b = -3.26$, $SE = 1.14$, $p = .004$, $q_{\text{FDR}} = .030$). In the univariate model, Conscientiousness remained the strongest predictor ($b = -2.51$, $p = .009$, $q = .065$), marginally surviving FDR at the liberal $q < .10$ threshold. Openness showed a positive trend toward larger shifts ($b = 1.39$, $p = .096$) but did not survive correction. No other moderator approached significance in either model.

3.6 Hierarchical Importance Decomposition

The full-model cross-validated R^2 for the susceptibility index was effectively zero (−0.0005), confirming that aggregate profile-level prediction is negligible at this sample size. Relative importance percentages, which normalize marginal contributions to sum to 100%, attribute 32.0% to Big Five traits overall and 28.8% to demographics. Within Big Five, Conscientiousness alone accounts for 23.1%, followed distantly by Extraversion (7.1%), Openness (3.8%), Neuroticism (3.0%), and Agreeableness (2.2%). Supplementary Figure S5 visualizes this hierarchical decomposition; Figure 3 shows the individual-feature weight distribution.

Figure 1: Full factorial study design for Run 8.

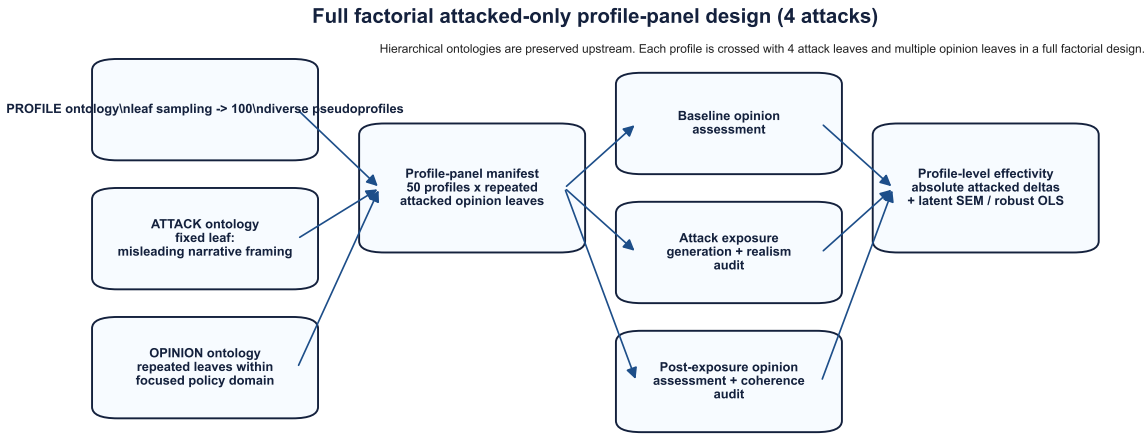


Figure 2: Absolute attacked opinion shift by opinion leaf.

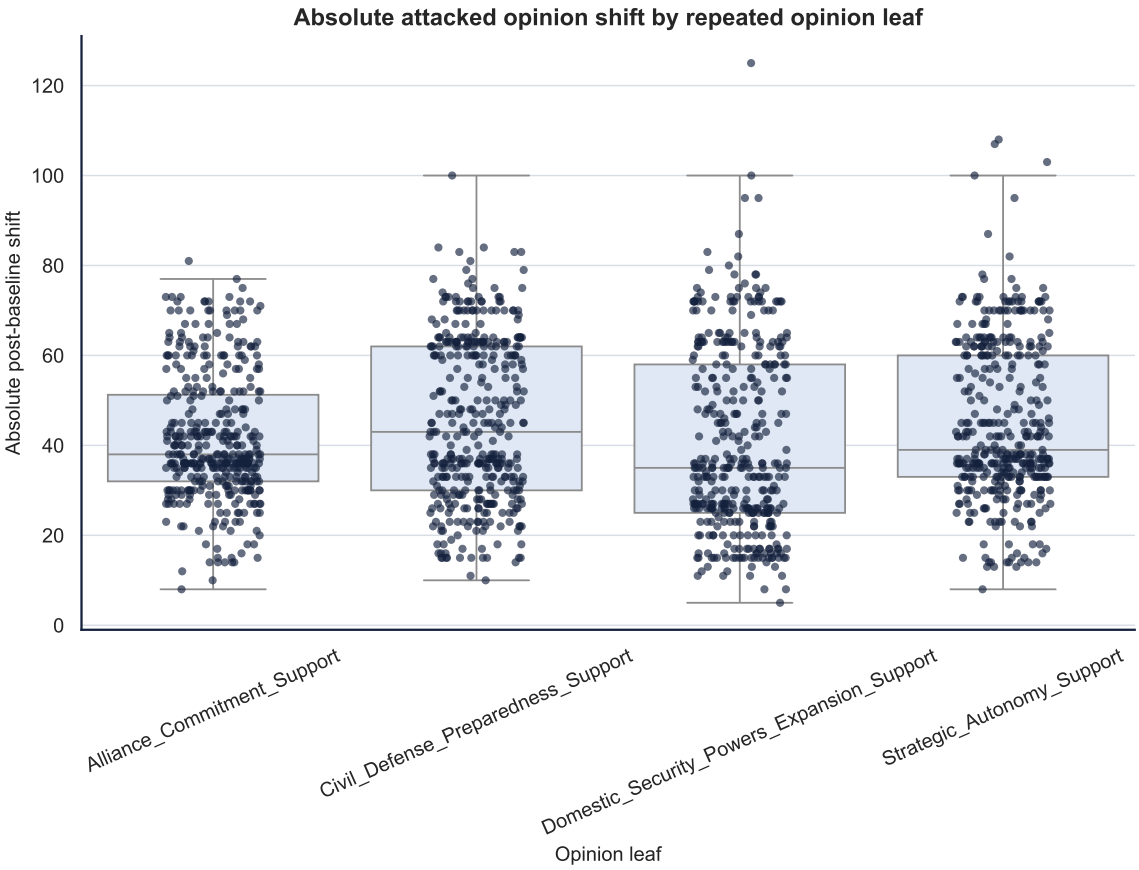


Figure 3: Normalized susceptibility weights across profile moderators.

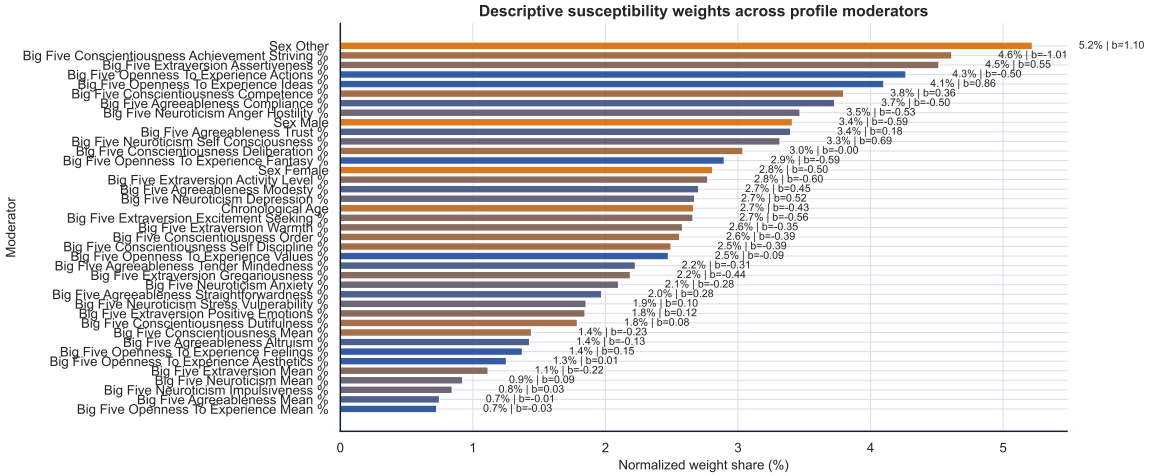
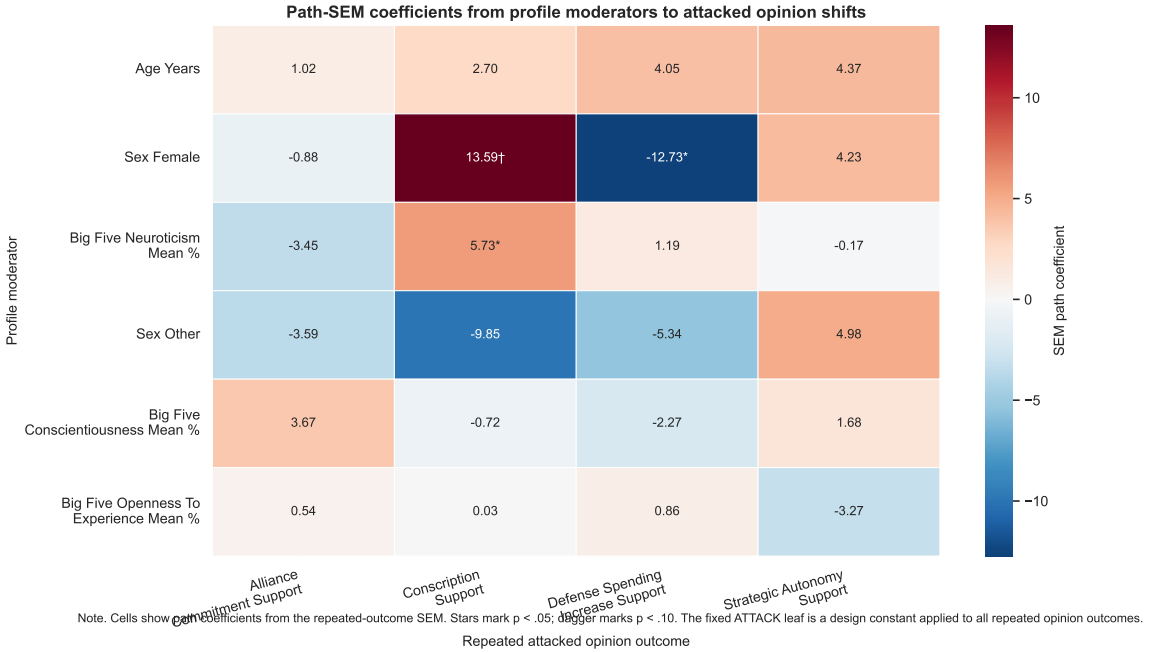


Figure 4: Repeated-outcome path-SEM coefficients across attacked opinion indicators.



Note. Cells show path coefficients from the repeated-outcome SEM. Stars mark p < .05; dagger marks p < .10. The fixed ATTACK leaf is a design constant applied to all repeated opinion outcomes.

Table 1: Pilot run design and configuration for the attacked-only profile-panel study.

Parameter	Value
Run ID	run_8
Paper title	PILOT: Inter-individual Differences in Susceptibility to Cyber-manipulation: A Multi-agent Simulation Approach with High-dimension State Space of Political Opinions
Profiles	100
Attacked rows	1600
Attack leaves	4 (Misleading_Narrative_Framing, Fear_Appeal_Scapegoating_Post, Astroturf_Comment_Wave, Pseudo_Expert_Authority_Cue)
Opinion domain	Defense_and_National_Security
Repeated opinion leaves	4
Design	Full factorial (100 x 4 x 4)
Model	mistralai/mistral-small-3.2-24b-instruct

Note. Full factorial design: each profile is crossed with 4 attack leaves and 4 opinion leaves. All rows are attacked; there is no no-attack control condition.

Table 2: Attacked effectivity descriptive statistics by repeated opinion leaf.

opinion_leaf_label	n_rows	mean_baseline	mean_post	mean_signed_delta	mean_abs_delta	sd_abs_delta	mean_adv_eff	sd_adv_eff
Civil_Defense_Preparedness_Support	400	351.842	388.480	36.638	45.443	18.056	-36.638	32.384
Strategic_Autonomy_Support	400	416.065	446.045	29.980	45.035	17.664	29.980	37.965
Alliance_Commitment_Support	400	447.688	459.685	11.998	41.562	14.241	-11.998	42.265
Domestic_Security_Powers_Expansion_Support	400	192.702	232.012	39.310	40.595	20.435	-39.310	22.810

Note. Descriptive statistics aggregated across 4 attack vectors per opinion leaf. Adversarial effectivity weights shift by the per-leaf adversarial goal direction.

Table 3: Profile moderators of attacked effectivity: controlled contrasts, SEM paths, and descriptive susceptibility weights.

Moderator	Role	Controlled mean b	Controlled p	Ridge mean b	Weight %	SEM mean b	SEM mean |b|	SEM min p	SEM paths
Big Five Conscientiousness Mean %	core	-2.511	0.009			-2.470	3.179	0.006	4.000
Big Five Neuroticism Mean %	core	-0.026	0.979			0.473	0.682	0.374	4.000
Sex Other	core	0.622	0.777	1.081	5.221	1.246	4.942	0.075	4.000
Sex Female	core	0.413	0.833	-0.696	2.809	1.400	4.708	0.095	4.000
Big Five Openness To Experience Mean %	core	1.395	0.096			0.899	1.478	0.179	4.000
Big Five Extraversion Mean %	exploratory	-0.986	0.340						
Big Five Agreeableness Mean %	exploratory	-0.277	0.735						

Note. Controlled coefficients summarize moderator associations with mean attacked effectivity. SEM columns summarize the repeated-outcome path model across the attacked opinion leaves. Weight percentages come from the target-conditional regularized aggregation used to compute the post hoc susceptibility index.

4 Discussion

4.1 Attack–Opinion Context Dominates Profile-Level Variance

The near-zero ICC values—0.052 for absolute shift and ≈ 0 for adversarial effectivity—establish that the combination of attack vector and opinion leaf is the primary determinant of outcome variance. Stable inter-individual differences explain a fraction of the total. This finding aligns with the broader persuasion literature showing that even psychologically targeted content produces modest average individual-level effects (Matz et al., 2017), and it constrains the practical utility of any profile-based susceptibility classification: the same individual may resist one attack on one issue and be substantially moved by a different combination.

4.2 Conscientiousness as the Dominant Moderator

Conscientiousness emerged as the most consistent predictor of reduced adversarial effectivity across the multivariate model, univariate comparison, SEM leaf-specific paths, and the hierarchical importance decomposition—the only moderator surviving FDR correction and accounting for nearly a quarter of relative importance. The negative sign is interpretable: Conscientiousness, characterized by deliberation, goal-directed self-regulation, and preference for carefully vetted information, plausibly confers a dispositional buffer against rapid adversarial opinion displacement. This is consistent with evidence that analytic cognitive style is associated with lower susceptibility to partisan misinformation and with greater ability to distinguish accurate from inaccurate information under time pressure (Pennycook & Rand, 2019, 2021). Conscientiousness may function as a proxy for reflective processing in the present pipeline, though this interpretation is necessarily tentative given the synthetic nature of the respondents.

4.3 Simulation as Conditional Measurement, Not Scalar Judgement

A critical consideration for interpreting these results is what the simulation pipeline produces relative to the alternatives. Direct one-shot elicitation—prompting an LLM to rate how susceptible a profile is—would yield a single scalar judgment compressing all person, attack, and issue information into an opaque response. The simulation pipeline instead produces a structured measurement system: 16 task-specific conditional susceptibility estimates (one per attack–opinion combination), repeated opinion indicators suitable for SEM, and an auditable staging of baseline, exposure, and post-exposure states. This object can be compared to human experimental data in ways that a scalar cannot—through criterion validity against observed opinion shifts, convergent validity with psychometric scales, and stability testing under prompt and seed variation. The present results should therefore be understood not as a claim that the model correctly ranks real individuals by susceptibility, but as evidence that the architecture produces internally consistent, analytically decomposable estimates of the right functional form: susceptibility as a conditional, task-specific, moderator-weighted response surface rather than an intrinsic person property.

4.4 Directional Asymmetry and Opinion-Specific Patterns

The finding that only Strategic Autonomy Support showed positive mean adversarial effectivity—while Civil Defense and Domestic Security opinions moved consistently against the intended adversarial direction—suggests that opinion-leaf characteristics interact strongly

with the simulation's internal opinion model. The directional asymmetry is consistent with psychological reactance: opinions that are framed in direct conflict with a threat-focused national-security context may generate motivated resistance rather than compliance (Roozenbeek et al., 2022). However, whether this pattern reflects genuine psychological dynamics or artifacts of the LLM's training distribution cannot be determined from the simulation alone, reinforcing the interpretive constraint that findings describe properties of the simulation architecture rather than a human population.

4.5 Limitations

The profiles are synthetic and the full-model cross-validated R^2 is near zero, confirming that overall profile-level prediction is weak at pilot scale. Standard errors do not account for the clustered structure of 16 observations per profile; future work should apply cluster-robust inference or profile-level bootstrap resampling (Cameron et al., 2008). The near-perfect SEM fit (CFI = 1.000) is expected in a near-saturated four-indicator, 100-observation model and does not constitute independent construct validation. The susceptibility index is descriptive because it derives from the same model whose moderator structure it summarizes. Criterion validity against observed human opinion shifts under analogous attack conditions remains the necessary next step.

5 Conclusion

This pilot demonstrates that a full factorial ontology-driven simulation—100 profiles, 4 attack vectors, 4 opinion leaves—produces an internally consistent and analytically decomposable measurement architecture for studying susceptibility to adversarial opinion manipulation. The five methodological contributions relative to earlier single-attack pilots are: (1) a four-vector attack taxonomy spanning cognitive, emotional, social-proof, and authority-based persuasion mechanisms; (2) adversarial effectivity as a directional primary outcome distinguishing successful manipulation from backfire; (3) intraclass correlation to establish the variance-partitioning ceiling for profile-level moderation; (4) Benjamini–Hochberg FDR correction for the moderator screening battery; (5) a hierarchical leave-one-group-out importance decomposition for the conditional susceptibility index. The dominant empirical finding—Conscientiousness as the most consistent moderator of reduced susceptibility across multiple analytic lenses—is theoretically coherent with deliberative processing accounts of resistance to persuasion. Substantive claims about real populations require validation against human experimental data; the present pilot provides the methodological infrastructure for that comparison.

A Supplementary Materials

Figure 5: Baseline versus post-attack opinion scores across all 1,600 scenarios.

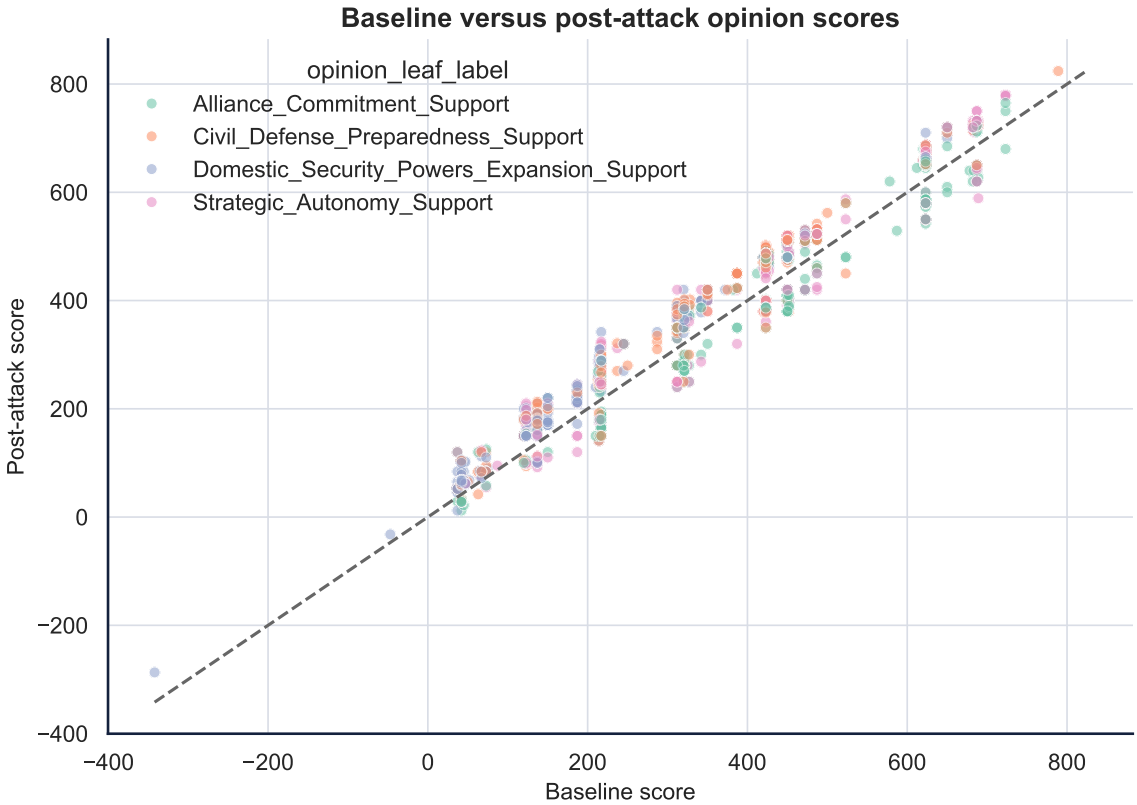


Figure 6: Per-profile attack effectivity heatmap (profiles ordered by susceptibility index).

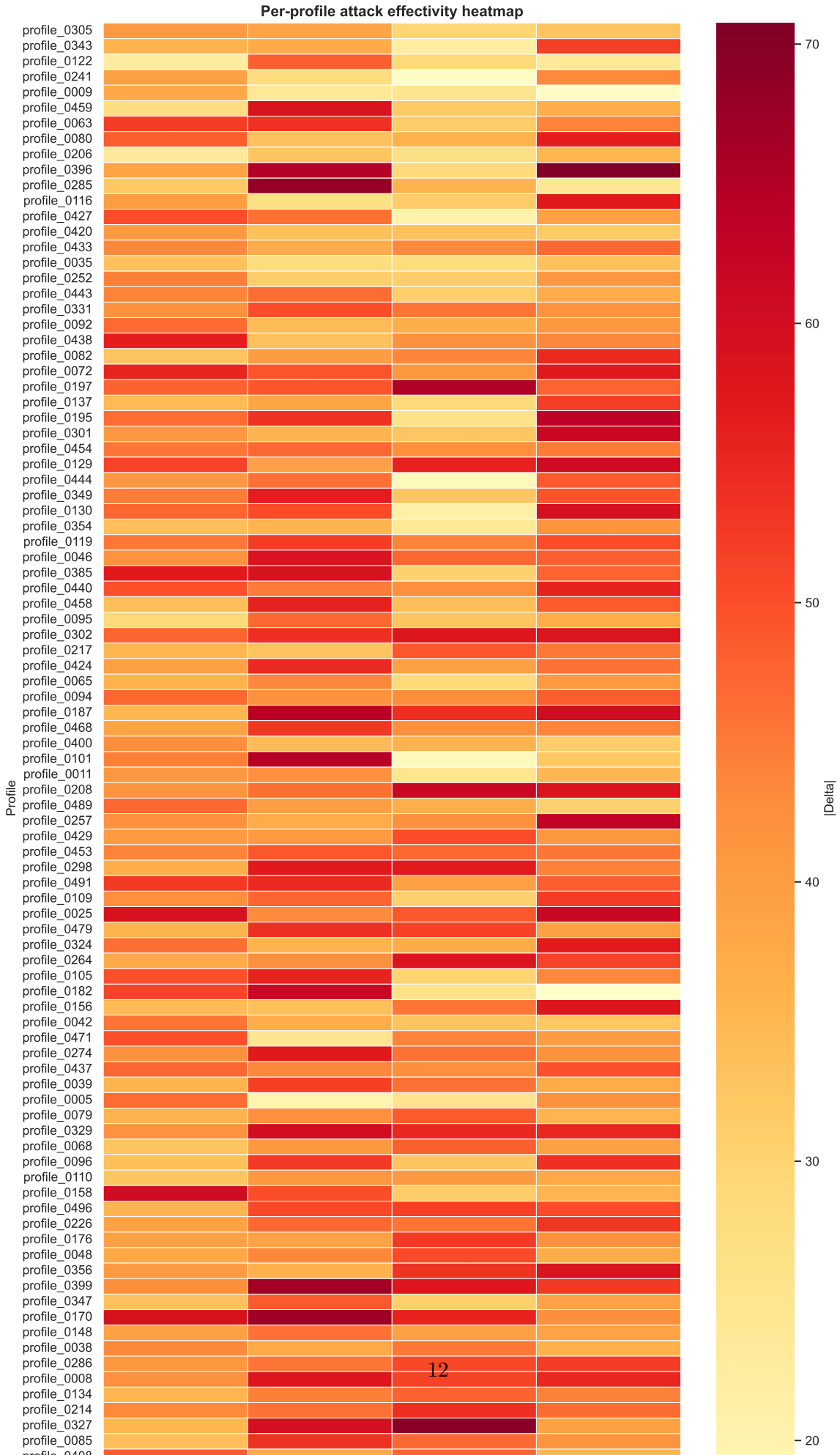


Figure 7: Distribution of the post hoc empirical susceptibility index across 100 profiles.

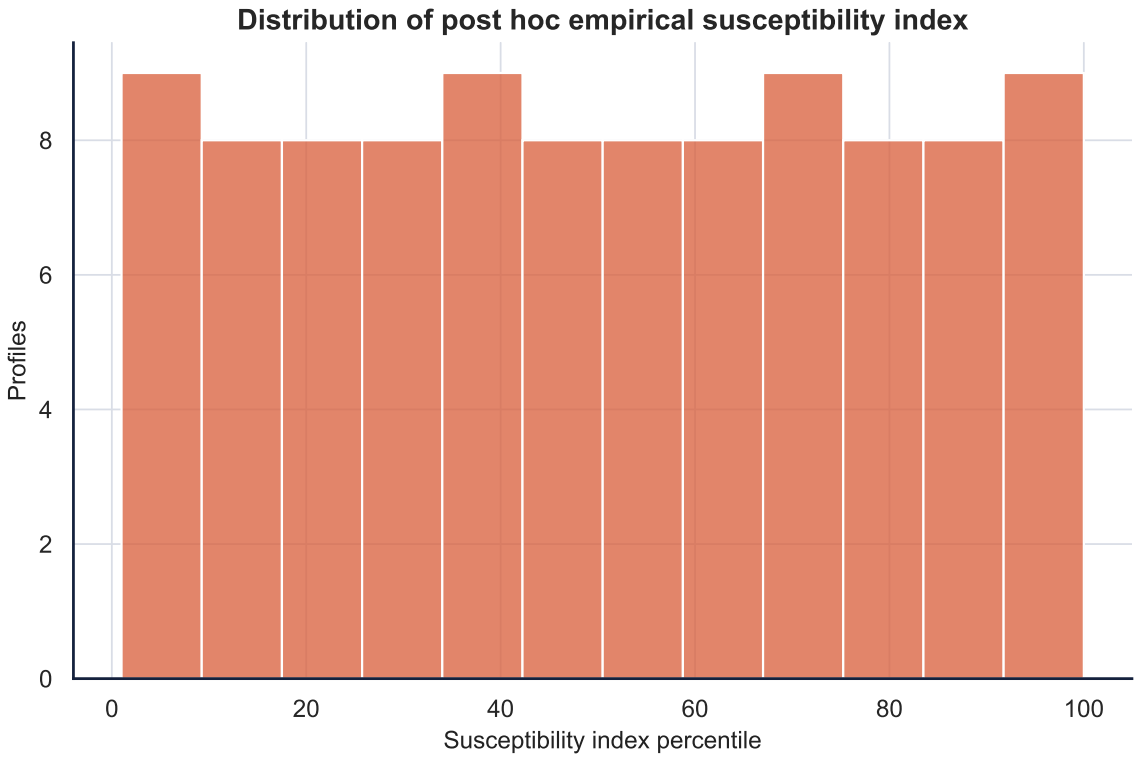


Figure 8: Adversarial effectivity by attack vector.

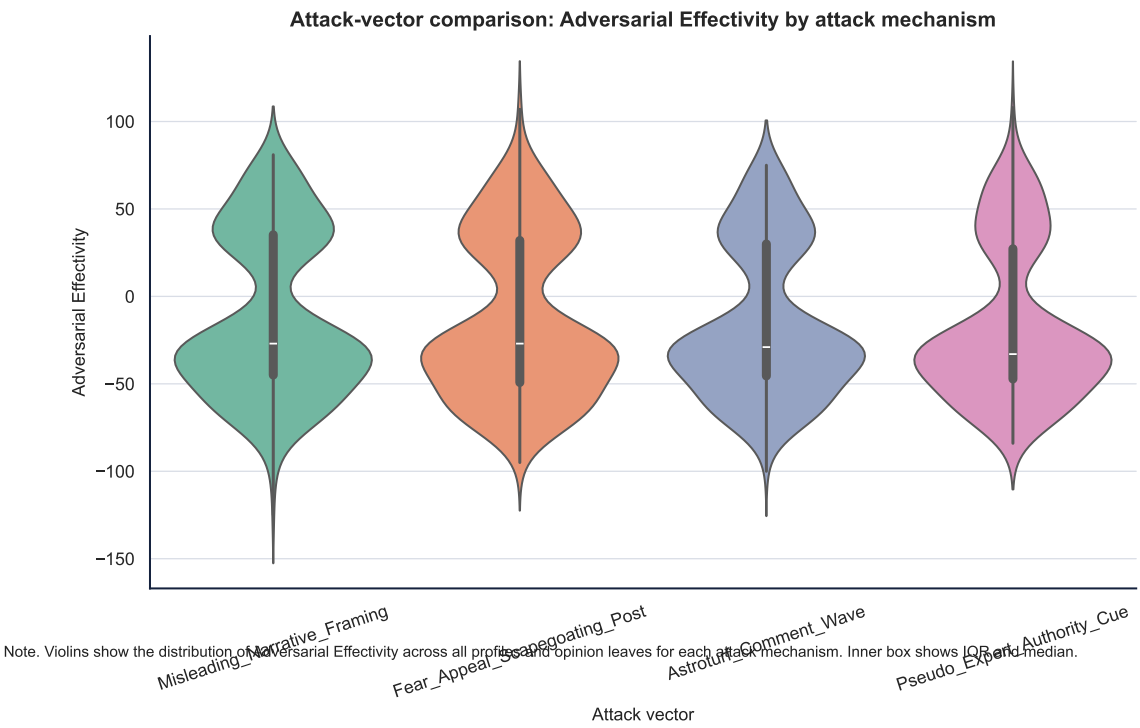


Figure 9: Hierarchical feature importance in the conditional susceptibility model.

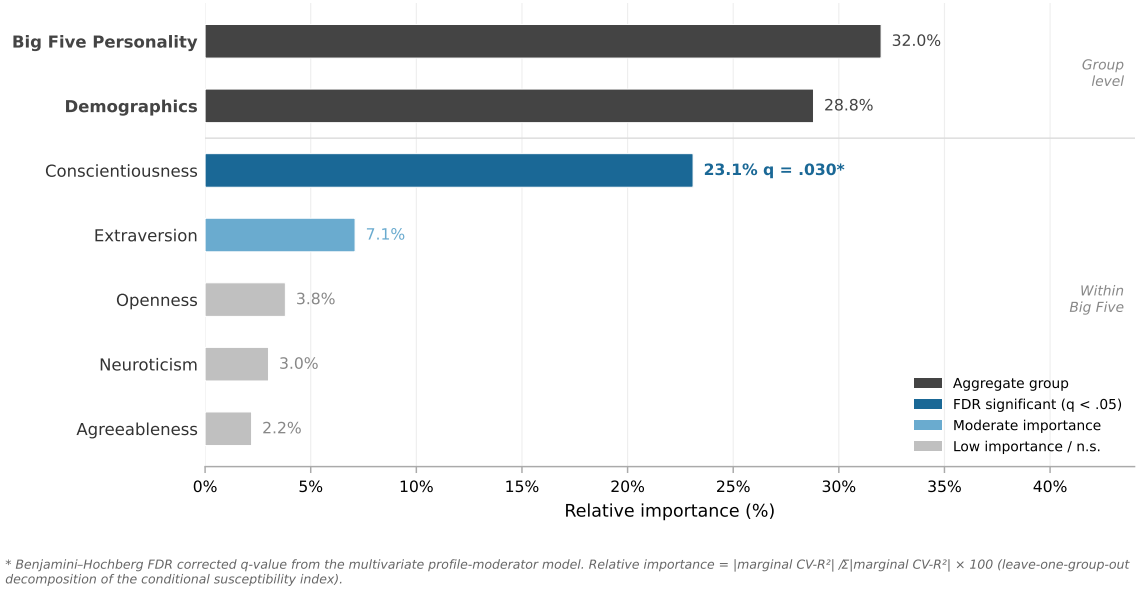


Table 4: Ontology leaves used in run_8.

Ontology	Role	Leaf
ATTACK	Factorial attack leaf	ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Framing
ATTACK	Factorial attack leaf	ATTACK_VECTORS > Manipulative_Persuasion > Fear_Appeal_Scapegoating_Post
ATTACK	Factorial attack leaf	ATTACK_VECTORS > Coordinated_Amplification > Astroturf_Comment_Wave
ATTACK	Factorial attack leaf	ATTACK_VECTORS > Manipulative_Persuasion > Pseudo_Expert_Authority_Cue
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Defense_and_National_Security > Alliance_Commitment_Support
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Defense_and_National_Security > Civil_Defense_Preparedness_Support
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Defense_and_National_Security > Domestic_Security_Powers_Expansion_Support
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Defense_and_National_Security > Strategic_Autonomy_Support

Note. Only ontology leaves are sampled for estimation. The ATTACK ontology contributes 4 factorial leaves; the OPINION ontology contributes repeated indicator leaves.

Table 5: Core and exploratory profile moderator comparison.

Moderator	Role	Multivariate b	Multivariate p	Univariate b	Univariate p	Ontology group	Normalized weight %
Big Five Conscientiousness Mean %	core	-3.258	0.004	-2.511	0.009	Personality: Conscientiousness	1.443
Big Five Neuroticism Mean %	core	1.273	0.261	-0.026	0.979	Personality: Neuroticism	0.923
Sex Other	core	2.395	0.357	0.622	0.777	Demographics: Sex	5.221
Sex Female	core	2.017	0.368	0.413	0.833	Demographics: Sex	2.809
Big Five Openness To Experience Mean %	core	0.567	0.499	1.395	0.096	Personality: Openness	0.727
Big Five Extraversion Mean %	exploratory	-2.353	0.130	-0.986	0.340	Personality: Extraversion	1.115
Big Five Agreeableness Mean %	exploratory	-0.422	0.602	-0.277	0.735	Personality: Agreeableness	0.748

Note. Core terms are entered into the latent SEM or primary multivariate profile model. Normalized weight percentages summarize each moderator's share of the total fitted importance after accounting for the moderator's observed variability.

Table 6: Assumption register and peer-review risk register.

Type	Item	Status	Mitigation
Assumption	Pilot sample size is sufficient for stable moderation estimates.	ok	Treat estimates as exploratory and scale scenario count in later runs.
Assumption	Attack-only design is intentional and aligned with the moderation question.	ok	Interpret results as moderation among attacked individuals, not as a no-attack counterfactual contrast.
Assumption	Opinion measurement has enough numeric resolution.	risk	Prompt enforces high-resolution scores in [-1000,1000].
Assumption	Attack text realism is adequate.	ok	Self-supervised realism reviewer + rewrite loop in Stage 03.
Assumption	SEM converges with acceptable fit.	ok	Use robust OLS supplement, reduce heterogeneity, and increase scenario count before substantive interpretation.
Assumption	LLM outputs are reproducible and auditable.	ok	Persist all prompts/outputs in provenance/raw_llm.
Critique	Synthetic LLM agents may not represent human cognition or causal response behavior.	addressed	Added transparent assumption register and explicit exploratory caveat in reports.
Critique	Adversarial content may be unrealistic or too generic.	addressed	Added realism review agent with rewrite loop and heuristic checks.
Critique	Model dependence on a single LLM could bias outputs.	addressed	Model is CLI-configurable and run metadata captures exact model for replication.
Critique	Insufficient sample size inflates uncertainty and may destabilize SEM.	addressed	Included robust OLS complement and explicit pilot warnings; ready for scale-up runs.
Critique	Attack-only designs cannot estimate the incremental effect of exposure relative to a no-attack counterfactual.	addressed	The report states explicitly that the attacked-only design estimates heterogeneity of post-minus-baseline response among attacked individuals only.

Note. The pilot is methodological and exploratory. Risks are surfaced explicitly so later scale-up runs can target the current design bottlenecks.

Table 7: Reproducibility manifest for run_8.

Field	Value
run_id	run_8
paper_title	PILOT: Inter-individual Differences in Susceptibility to Cyber-manipulation: A Multi-agent Simulation Approach with High-dimension State Space of Political Opinions
output_root	[project root]/evaluation/run_8
report_root	[project root]/research_report/report
report_assets_root	[project root]/research_report/assets
n_scenarios	10
n_profiles	100
seed	88
attack_ratio	1.000
attack_leaf	ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Framing
attack_leaves	Misleading_Narrative_Framing, Fear_Appeal_Scapegoating_Post, Astroturf_Comment_Wave, Pseudo_Expert_Authority_Cue
focus_opinion_domain	Defense_and_National_Security
max_opinion_leaves	4
profile_candidate_multiplier	5
primary_moderator	posthoc_profile_susceptibility_index
bootstrap_samples	800
use_test_ontology	True
ontology_root	[project root]/src/backend/ontology/separate/test
openrouter_model	mistralai/mistral-small-3.2-24b-instruct
temperature	0.150
max_repair_iter	2
profile_generation_mode	deterministic
self_supervise_attack_realism	True
realism_threshold	0.760
self_supervise_opinion_coherence	True
coherence_threshold	0.760
generate_visuals	True
export_static_figures	True
build_report	True
resume_from_stage	01
stop_after_stage	09
save_raw_llm	True
timeout_sec	90
max_concurrency	20
run_stage_checks	False
stage_check_scenarios	3

Note. The manifest captures the full pipeline configuration used to generate the pilot outputs and manuscript assets.

Table 8: Leaf-specific path-SEM coefficients.

Outcome leaf	Moderator	SEM b	SE	p
Adversarial Delta Indicator Alliance Commitment Support	Big Five Neuroticism Mean %	-0.417	2.291	0.856
Adversarial Delta Indicator Alliance Commitment Support	Big Five Openness To Experience Mean %	2.954	2.201	0.179
Adversarial Delta Indicator Alliance Commitment Support	Big Five Conscientiousness Mean %	-6.758	2.448	0.006
Adversarial Delta Indicator Alliance Commitment Support	Sex Female	4.457	5.300	0.400
Adversarial Delta Indicator Alliance Commitment Support	Sex Other	-1.042	5.543	0.851
Adversarial Delta Indicator Alliance Commitment Support	Mean Baseline Abs Score	-1.116	2.831	0.694
Adversarial Delta Indicator Alliance Commitment Support	Mean Exposure Quality Score	-2.318	2.252	0.303
Adversarial Delta Indicator Civil Defense Preparedness Support	Big Five Neuroticism Mean %	1.524	1.713	0.374
Adversarial Delta Indicator Civil Defense Preparedness Support	Big Five Openness To Experience Mean %	-0.589	1.645	0.720
Adversarial Delta Indicator Civil Defense Preparedness Support	Big Five Conscientiousness Mean %	-4.541	1.830	0.013
Adversarial Delta Indicator Civil Defense Preparedness Support	Sex Female	-6.616	3.962	0.095
Adversarial Delta Indicator Civil Defense Preparedness Support	Sex Other	-6.348	4.144	0.126
Adversarial Delta Indicator Civil Defense Preparedness Support	Mean Baseline Abs Score	-1.202	2.117	0.570
Adversarial Delta Indicator Civil Defense Preparedness Support	Mean Exposure Quality Score	-1.285	1.684	0.445
Adversarial Delta Indicator Domestic Security Powers Expansion Support	Big Five Neuroticism Mean %	0.268	1.170	0.819
Adversarial Delta Indicator Domestic Security Powers Expansion Support	Big Five Openness To Experience Mean %	-0.568	1.124	0.613
Adversarial Delta Indicator Domestic Security Powers Expansion Support	Big Five Conscientiousness Mean %	0.256	1.250	0.838
Adversarial Delta Indicator Domestic Security Powers Expansion Support	Sex Female	2.595	2.706	0.338
Adversarial Delta Indicator Domestic Security Powers Expansion Support	Sex Other	4.257	2.830	0.133
Adversarial Delta Indicator Domestic Security Powers Expansion Support	Mean Baseline Abs Score	-6.528	1.445	0.000
Adversarial Delta Indicator Domestic Security Powers Expansion Support	Mean Exposure Quality Score	0.382	1.150	0.740
Adversarial Delta Indicator Strategic Autonomy Support	Big Five Neuroticism Mean %	0.519	1.884	0.783
Adversarial Delta Indicator Strategic Autonomy Support	Big Five Openness To Experience Mean %	1.800	1.809	0.320
Adversarial Delta Indicator Strategic Autonomy Support	Big Five Conscientiousness Mean %	1.161	2.013	0.564
Adversarial Delta Indicator Strategic Autonomy Support	Sex Female	5.162	4.358	0.236
Adversarial Delta Indicator Strategic Autonomy Support	Sex Other	8.119	4.557	0.075
Adversarial Delta Indicator Strategic Autonomy Support	Mean Baseline Abs Score	0.204	2.328	0.930
Adversarial Delta Indicator Strategic Autonomy Support	Mean Exposure Quality Score	4.800	1.852	0.010

Note. Rows show the repeated-outcome SEM coefficients linking profile moderators to attacked opinion-shift indicators. Each opinion indicator is the adversarial delta averaged across the four attack vectors; attack-specific variation is examined in Supplementary Figure S4.

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